

Tool Disclosure

PyMentor Personalized Python Tutor

Seif Mohamed (202301506) | Patrick Saweris (202301486)

Tool / data	Purpose and architectural role
Python 3.9+	Implementation language and subject taught.
LangGraph	Required graph orchestration: state, nodes, edges, and routing.
Pydantic	Validates structured response and source schemas.
SQLite	Persists sessions, profiles, quiz history, and misconceptions.
Requests	Calls Groq and local Ollama HTTP endpoints.
Groq API	Supported optional hosted provider; no key used in final metrics.
Ollama / qwen3:4b	Final 32-case, LLM-judge, and RAGAS inference provider.
Streamlit	Interactive live-demo interface with visible sources and flags.
RAGAS	Final faithfulness and retrieval evaluation.
Pytest	Deterministic tests for retrieval, memory, and guardrails.
Synthetic Python notes	Controlled CSAI 106-level grounding corpus.
Synthetic test conversations	32 cases across personas, edge cases, and attacks.
OpenAI Codex	Assisted guideline analysis, implementation, documentation, and testing; all work remains the team's responsibility.
Course lectures and labs	Guided prompting, RAG, memory, LangGraph, tools, safety, and evaluation choices.

Credential statement: No API key is included in the repository. Secrets are loaded from an ignored .env file. Synthetic and AI-assisted materials are disclosed above.